Final Exam

Fundamentals of One- and Two-Way ANOVA

Question 1 4 / 4 points

You are performing an experiment to compare five treatments using a completely randomized design. The number of experimental units per treatment group is 9. You would like to construct simultaneous 95% confidence intervals for different contrasts involving the treatment means $\sum_{i=1}^{5} c_i \mu_i$. Each confidence interval is of the form $\sum_{i=1}^{5} c_i \hat{\mu}_i \pm w \times SE(\sum_{i=1}^{5} c_i \hat{\mu}_i)$, where w depends on the procedure being used to control the family-wise error rate for the intervals.

Consider the following four tasks:

- Using the Bonferroni procedure to construct confidence intervals for all treatment means as well as all pairwise comparisons between treatment means.
- Using the Scheffe procedure to construct confidence intervals for all treatment means as well as all pairwise comparisons between treatment means.
- Using the Tukey procedure to construct confidence intervals for all pairwise comparisons between treatment means.
- Using the Dunnett procedure to construct confidence intervals for comparisons between each of the last four treatments with the first treatment.

In the four boxes below, provide the value for w that would be used when constructing the simultaneous confidence intervals for these four scenarios. Please provide your answers in the order that the scenarios were presented above. **Round any intermediate calculations to four decimal places, but round your *final answers to two decimal places* in the boxes below.**

Answer for blank # 1: 3.12 ✓(25 %)

Answer for blank # 2: 3.23 ✓(25 %)

Answer for blank # 3: 2.86 ✓(25 %)

Answer for blank # 4: 2.54 ✓(25 %)

Question 2 1 / 1 point

You are interested in studying the effect of two factors, diet and exercise, on weight. To do this, you adopt a completely randomized design. Some individuals will be asked to diet while others will not. Likewise, some individuals will be asked to exercise while others will not. You randomly select 16 individuals to participate in the study and then randomly assign 4 individuals to each of the four combinations of diet and exercise. You then monitor participants for one month and record the change in weight (final weight minus beginning weight) for each participant at the end of the study.

You analyze the data from this experiment using the following model:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where α_i refers to the effect of diet i, β_j refers to the effect of exercise j, and $(\alpha\beta)_{ij}$ refers to the effect of an interaction between diet i and exercise j. Furthermore, $i = 1$ indicates no diet. Likewise, $j = 1$ indicates no exercise.

Source	**Sums of Squares**	**Degrees of Freedom**
Diet	300	1
Exercise	800	1
Diet*Exercise	1200	1
Error	700	12
Total	2900	15

Use this information to compute the **width** of the 95% confidence interval for the expected difference in weight change between the group that diets and exercises and the group that doesn't diet but still exercises. (The width is defined as the upper bound of the interval minus the lower bound).

Round any intermediate answers to four decimal places, but round your *final answer to two decimal places*.

Answer:

23.53 ✓

Question 3 1 / 1 point

You are investigating the effectiveness of different study methods on student performance for a standardized test. You randomly select 24 students to participate in your study. You randomly assign 6 students to prepare using flashcards, 6 students to prepare using online practice tests, 6 students to prepare using group study sessions, and 6 to prepare without any additional materials to serve as a control group. After allowing one week for students to prepare, the standardized test is administered and a score between 0 and 100 is recorded for each student.

Below is a table containing the scores for each student that participated in the experiment:

Method	Response
Flashcards	78
Flashcards	82
Flashcards	81
Flashcards	75
Flashcards	87
Flashcards	78
Online Tests	83
Online Tests	85
Online Tests	79
Online Tests	80
Online Tests	84
Online Tests	82
Group Study	85
Group Study	86
Group Study	90
Group Study	87
Group Study	91
Group Study	84
No Materials	78
No Materials	75
No Materials	68
No Materials	80
No Materials	76
No Materials	76

You use the following one-way ANOVA model for these data: $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ where $\varepsilon_{ij} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$. Compute the upper bound for a 95% one-sided confidence interval for σ^2. **Round any intermediate calculations to four decimal places, but round your *final answer to two decimal places.***

Answer:

21.75 ✓

Block Designs

Fertilizer Experiment

In an agricultural study to evaluate the effect of fertilizer types and irrigation levels on tomato yield, a randomized complete block design (RCBD) was implemented where the blocks represent different plots of soil. The experiment considered two treatment factors: fertilizer type (organic, inorganic, and mixed) and irrigation level (low and high). Each treatment combination was randomly assigned within each block (plot of soil), resulting in a block size of 6. The study aimed to measure tomato yield across these treatments, allowing for analysis of the main effects of fertilizer and irrigation, their interaction, and the variability due to the plots of soil (blocks).

Use this information to answer the following two questions.

Question 4 1 / 1 point

You would like to construct simultaneous 95% confidence intervals for all pairwise comparisons of fertilizer types (organic, inorganic, and mixed) using Tukey's procedure. How many blocks are necessary to use so that the width of these intervals is less than 30? (For this problem, assume the MSE is 500).

- ○ 12
- ○ 5
- ○ 4
- ✓ ◉ 13

Question 5

1 / 1 point

Assume that the number of blocks used for this experiment is 4. You analyze the data from this experiment and obtain the following ANOVA table:

Source	Sums of Squares
Block	400
Fertilizer	1200
Irrigation	550
Fertilizer*Irrigation	460
Error	2500

What is the value of the F-statistic that would be used to test the hypothesis of no effect from the type of fertilizer?

- ○ 3.84
- ○ 2.56
- ○ 2.40
- ✔ ◉ 3.60

Question 6

1 / 1 point

Consider an incomplete block design to compare 6 treatments in blocks of size 4.

What is the minimum number of blocks needed so that the design is a **balanced** incomplete block design?

- ✔ ◉ 15
- ○ 5
- ○ 3
- ○ Cannot be determined

Random Effects Model

Employee Training Program Experiment

A study is conducted to evaluate the effectiveness of a training program on employee productivity. Five departments are randomly selected among all possible departments and 12 individuals are selected at random from each department. Then, four of these 12 individuals are randomly assigned to training program A, four to training program B, and four to training program C. Productivity scores are collected before and after the start of each program and the difference in scores are used as the response variable. The following mixed effects model is used for these, considering the training program (α) as a fixed effect and department (B) as a random effect:

$$y_{ijk} = \mu + \alpha_i + B_j + (\alpha B)_{ij} + \varepsilon_{ijk}.$$

Use this information to answer the following three questions.

Question 7 1 / 1 point

Which of the following options gives the correct expected mean squares for the training program factor in this experiment?

- ○ $\sigma^2 + 4\sigma^2_{AB} + 20\sigma^2_A$
- ○ $\sigma^2 + Q(A)$
- ○ $\sigma^2 + 20\sigma^2_A$
- ✓ ◉ $\sigma^2 + 4\sigma^2_{AB} + Q(A)$

Question 8 1 / 1 point

Assume that the data for this experiment are as follows:

```
data prod;
input program $ department $ score;
datalines;
A Dept1 2
A Dept1 3
A Dept1 4
A Dept1 4
B Dept1 6
B Dept1 5
B Dept1 4
B Dept1 6
C Dept1 3
C Dept1 2
C Dept1 3
C Dept1 4
A Dept2 1
A Dept2 2
A Dept2 5
A Dept2 4
B Dept2 5
B Dept2 6
B Dept2 7
B Dept2 8
C Dept2 4
C Dept2 3
C Dept2 4
C Dept2 5
A Dept3 3
A Dept3 4
A Dept3 5
A Dept3 6
B Dept3 5
B Dept3 8
B Dept3 7
B Dept3 6
C Dept3 4
C Dept3 5
C Dept3 6
C Dept3 5
A Dept4 2
A Dept4 3
A Dept4 4
A Dept4 5
B Dept4 7
B Dept4 6
B Dept4 7
B Dept4 8
C Dept4 4
C Dept4 2
C Dept4 4
C Dept4 5
A Dept5 3
A Dept5 4
A Dept5 5
A Dept5 6
B Dept5 7
B Dept5 6
B Dept5 7
B Dept5 8
C Dept5 6
C Dept5 3
C Dept5 4
C Dept5 5
;
run;
```

You would like to use Tukey's procedure to obtain simultaneous 95% confidence intervals for all pairwise comparisons of the training programs. Which of the following options best describes the conclusion that can be drawn from these intervals?

- None of the programs are significantly different.
- Programs A and B are significantly different, as are programs A and C, but programs B and C are not significantly different.
- Only programs A and B are significantly different.
- Programs A and B are significantly different, as are programs B and C, but programs A and C are not significantly different.

Question 9 1 / 1 point

Consider the following SAS output:

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	112.2333333	8.0166667	7.40	<.0001
Error	45	48.7500000	1.0833333		
Corrected Total	59	160.9833333			

R-Square	Coeff Var	Root MSE	score Mean
0.697174	22.22419	1.040833	4.683333

Source	DF	Type I SS	Mean Square	F Value	Pr > F
program	2	84.23333333	42.11666667	38.88	<.0001
department	4	14.40000000	3.60000000	3.32	0.0181
program*department	8	13.60000000	1.70000000	1.57	0.1612

Source	DF	Type III SS	Mean Square	F Value	Pr > F
program	2	84.23333333	42.11666667	38.88	<.0001
department	4	14.40000000	3.60000000	3.32	0.0181
program*department	8	13.60000000	1.70000000	1.57	0.1612

What is the correct value of the F-statistic that would be used to test the hypothesis of no effect due to training program?

- ○ 38.88
- ○ 2.12
- ○ 3.32
- ✔ ◉ 24.77

Nested Models and Split-Plot Designs

Question 10 1 / 1 point

Consider the following table of expected mean squares for an experimental design with two factors, A and B (both random), where factor B is nested within factor A:

Source	Expected Mean Square
A	$\sigma^2 + 40\sigma_A^2 + 4\sigma_{B(A)}^2$
B(A)	$\sigma^2 + 4\sigma_{B(A)}^2$
Error	σ^2

Using this information, which of the following equations would be used to obtain the F-statistic for testing the hypothesis of no effect due to factor A (i.e., $\sigma_A^2 = 0$)?

- ○ $\frac{MS_A}{MS_{Error}}$
- ✓◉ $\frac{MS_A}{MS_{B(A)}}$
- ○ $\frac{MS_{B(A)}}{MS_{Error}}$
- ○ $\frac{MS_{B(A)}}{MS_A}$

Irrigation and Pesticide Experiment

In a large agricultural field, a study is conducted to evaluate the effects of irrigation method and pesticide type on crop yield using a split plot design. The field is divided into several blocks, with each block further divided into two main plots for the irrigation methods (drip and sprinkler irrigation). Each main plot is then subdivided into two subplots for the pesticide types (Pesticide A and Pesticide B), resulting in 4 subplots per block. Crop yield is then recording at the end of the growing season.

Use this information to answer the following two questions.

Question 11 1 / 1 point

Which of the following statements about the factors in this experiment is correct?

- ○ The whole plot factor is the pesticide type and the split-plot factor is the irrigation method.
- ○ Both factors would be considered split-plot factors and the whole plot simply serves as a blocking factor.
- ✓◉ The whole plot factor is the irrigation method and the split-plot factor is the pesticide type.
- ○ Both factors would be considered whole plot factors and the split-plots are simply there for replication of experimental units.

Question 12 — 1 / 1 point

Assume the data for the experiment are as follows:

```
data crop_yield;
input block irrigation pesticide yield;
datalines;
1 1 1 55
1 1 2 50
1 2 1 60
1 2 2 58
2 1 1 52
2 1 2 48
2 2 1 65
2 2 2 63
3 1 1 53
3 1 2 51
3 2 1 57
3 2 2 55
4 1 1 49
4 1 2 47
4 2 1 64
4 2 2 62
5 1 1 69
5 1 2 67
5 2 1 61
5 2 2 59
;
run;
```

What is the correct value of the F-statistic that would be used to test the hypothesis of no effect due to the irrigation method?

- 2.69
- 2.50
- 149.41
- 396.90

Additional Topics

Question 13 — 1 / 1 point

Which of the following tables represents an example of a Latin-Square design for a single treatment factor with four treatments labeled A-D (rows correspond to the first blocking factor and columns correspond to the second blocking factor)?

A	B	C	D
C	D	A	B
A	B	C	D
C	D	A	B

A	B	D	C
D	C	A	B
A	D	C	B
D	B	A	C

A	B	C	D
A	B	C	D
A	B	C	D
A	B	C	D

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

Question 14

1 / 1 point

You are interested in studying the effect of four different dietary supplements on weight gain in rats. There are two blocking factors, the cage in which a rat is kept as well as the time of day they are fed. There are four cages and four feeding times each day. Each cage contains four rats, one rat for each of the four feeding times.

The data for this Latin-Square design are as follows:

```
data latin_square;
input cage feeding_time supplement $ weight_gain;
datalines;
1 1 S1 12.5
1 2 S2 14.0
1 3 S3 13.0
1 4 S4 15.5
2 1 S2 13.5
2 2 S3 15.0
2 3 S4 16.5
2 4 S1 12.0
3 1 S3 14.5
3 2 S4 16.0
3 3 S1 12.5
3 4 S2 13.5
4 1 S4 15.5
4 2 S1 13.0
4 3 S2 14.0
4 4 S3 15.0
;
run;
```

Which of the following gives the F-statistic that would be used to test the hypothesis of no effect due to the dietary supplement?

- ✓ 18.90
- 22.91
- 6.73
- 0.70

Question 15

1 / 1 point

You are using a completely randomized design to compare different concentrations of a drug for a specific medication. You also have available to you two other useful sources of information about each individual in the study: **Their weight (x)** ***and*** **their age (z)**. You would like to include this information as covariates to improve the precision of your estimation of the effect due to the drug. Which of the following equations specifies the **most correct** ANCOVA model for this scenario?

- $y_{ij} = \mu + \tau_i + \beta(x_{ij} + z_{ij}) + \varepsilon_{ij}$
- $y_{ij} = \mu + \tau_i + \beta x_{ij} + \varepsilon_{ij}$
- ✓ $y_{ij} = \mu + \tau_i + \beta_1 x_{ij} + \beta_2 z_{ij} + \varepsilon_{ij}$
- $y_{ij} = \mu + \tau_i + \beta x_{ij} z_{ij} + \varepsilon_{ij}$